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LETTER

Attribution of the record-high 2023 SST using a deep-learning framework

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Abstract

The global-mean sea surface temperature (SST) reached a record high in 2023, exceeding the 2016 record by 0.14 °C. This unprecedented change in global-mean SST has major implications for our understanding of internal variability and the forced response in our changing climate. In this work, we use neural networks trained on simulated climate data to separate the contributions of internal variability and the forced response within observations. Performing attribution reveals that internal variability was responsible for +0.07 °C of the 2023 global mean SST, due to anomalously warm conditions in the Pacific, Atlantic, and Indian Ocean basins. Furthermore, these results provide a line of evidence for accelerated forced warming in recent years. Continued monitoring of the climate will be critical for understanding the drivers behind this unprecedented SST record.

1. Introduction

The global-mean sea surface temperature (SST) reached a record high in 2023, exceeding the previous record by 0.14 °C and the 1981–2010 average by more than 0.5 °C [1, 2]. Global-mean SST is a product of both externally forced change (e.g., human-caused warming) and variability internal to the climate system. This record was unprecedented and unpredicted: No other record had surpassed a previous record to such a large extent, and dynamical and statistical forecasting systems failed to capture such an extreme [3, 4]. Recent work has been aimed at understanding this record warmth. These studies suggest that while internal variability contributed to the high temperatures of 2023, the warming may also be linked to external forcings [5], such as reduced aerosol emissions due to recent changes in shipping regulations [6, 7], or an emerging low-cloud feedback [8].

The extent to which the record-high SST in 2023 is a product of internal variability or the forced response has major implications for our understanding of the climate system. An abrupt change in the forced response could indicate accelerating or nonlinear warming [9, 10], due to changes in SST patterns [e.g., 11] or the triggering of a climate tipping point [12]. Abrupt warming would result in major impacts on human and natural systems globally, including ecology [13, 14] and the economy [15], with further implications for human well-being and behavior [16]. Attribution of the 2023 global mean SST is also critical for driving discussions on climate change policy [3]. International agreements, such as the United Nations Paris Agreement, have established warming targets to limit the impacts of forced climate change [17, 18]. Actively monitoring the current proximity of the climate system to these warming targets requires immediate estimation of the forced response [19].

The Intergovernmental Panel on Climate Change (IPCC) uses the 20-year centered rolling average temperature to estimate the global warming magnitude within climate models [20]. This requires knowledge of the climate system a decade into the future, which is not available for real-world data and thus cannot be used to estimate the global warming magnitude in observations. Traditional methods, such as linear or fourth-order polynomial fits to historical observations [e.g., 21, 22], could be used to disentangle the forced and internal

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components in 2023, but they lack the ability to capture complex nonlinear behavior or abrupt changes in the forced response [e.g., the temperature response to the eruption of Mount Pinatubo; 23]. Hybrid methods that combine recent observations with decadal climate forecasts [e.g., 19] can also provide near-instantaneous estimates of the current global warming magnitude, but they rely on climate projections that do not include abrupt changes to future forcing [e.g., 24]. Fingerprinting methods [e.g., 25, 26] identify distinct spatial patterns of forced change within climate model simulations which can be projected onto observations to isolate the forced component of change. Since fingerprinting is not sensitive to the time-evolution of climate change, it is a natural fit for detecting sudden shifts in the forced response. However, simulated patterns of the forced response are inconsistent with recent observed SST trends, particularly in the tropical Pacific and the Southern Ocean [27, 28], and climate model-derived fingerprints are encoded with these same forced response biases.

In this work, we develop a deep learning method for attribution of the record-high 2023 SST, following recent use cases for deep learning for climate change attribution [29, 30]. A neural network is trained to separate the internal and forced components of annual-mean SST by learning patterns of forced change and internal variability from thousands of realizations of SST simulated by climate models. Once trained, the neural network uses these learned patterns to estimate the forced and internal components of observed SST. In the following, we will show that our deep learning approach provides skillful attribution for out-of-sample climate model data, and explore the extent to which the record-high SST in 2023 can be attributed to an abrupt change in the forced response or anomalously warm internal variability.

2. Data and methods

Our study uses a neural network to separate internal variability and the forced response for annual-mean SST. Following the framework and data provided by the ongoing Forced Component Estimation Statistical Methods Intercomparison Project [ForceSMIP; 31], our neural network leverages data from five climate model initial-condition large ensembles [32–36]. These large ensembles simulate the climate under historical forcings (1850–2014) and three future forcing scenarios (2015–2100) [37]. MIROC-ES2L (30 members) uses the SSP2-4.5 forcing scenario, CESM2 (50 members) uses the SSP3-7.0, and MIROC6 (50 members), CanESM5 (25 members), and MPI-ESM1-2-LR (30 members) use the SSP5-8.5. All climate model data is regridded to a $2.5^\circ \times 2.5^\circ$ resolution, and we select the years 1950–2022 for training. Since each large ensemble has at least 25 members, we ensure each climate model large ensemble is equally represented by using the first 17 members for training. The following eight members are used to explore how the neural network learns patterns of the forced response and internal variability (section S1).

In a single model initial condition large ensemble, the forced and internal components are known: the ensemble mean SST is the forced climate response while the residual is internal variability [38]. In this supervised learning task, we have the option to either predict the forced or internal component. We predict internal variability, calculating the forced response by subtraction, for several reasons. First, deep learning algorithms are prone to overfitting on small training sets, and predicting the internal component provides more distinct samples to train on. While our training set comprises 6205 samples (5 climate models $\times$ 17 members $\times$ 73 years), all with unique patterns of internal variability, there is only one forced response pattern for each climate model ensemble. This leaves only 365 unique maps of the forced response (5 climate models $\times$ 73 years), and fewer truly independent samples as the simulated forced response is highly autocorrelated.

Patterns of internal variability are also more complex than the patterns of the forced response [in fingerprinting methods, the forced response can be explained by just a few patterns while internal variability requires many; e.g., 39]. While internal variability and the forced response are two sides of the same coin–the neural network need only learn one to calculate the other–targeting internal variability pushes the network to learn the behavior of internal variability, not just the forced response (section S1). Climate models have been unable to capture the recent estimates of the forced response in observations [e.g., suppressed warming in the Eastern Tropical Pacific; 40] while simulations of internal variability have improved across model generations [41]. We hence expect neural networks that have learned the behavior of internal variability to generalize better to observations than neural networks that have simply learned the simulated pattern of the forced response.

Our prediction task is set up as follows: The neural network is given a map of simulated annual-mean SST as input and tasked to estimate the internal variability component at each gridpoint (figure 1). The input maps are preprocessed to remove differences in the mean temperature patterns between models following the procedure in figure 1(a) and section S2. To learn local and global patterns of internal variability and the forced response, the neural network architecture employs convolutional layers, fully-connected layers, and skip connections (figure 1(b)). The full neural network architecture and the hyperparameter tuning process are discussed in sections S3 and S4.

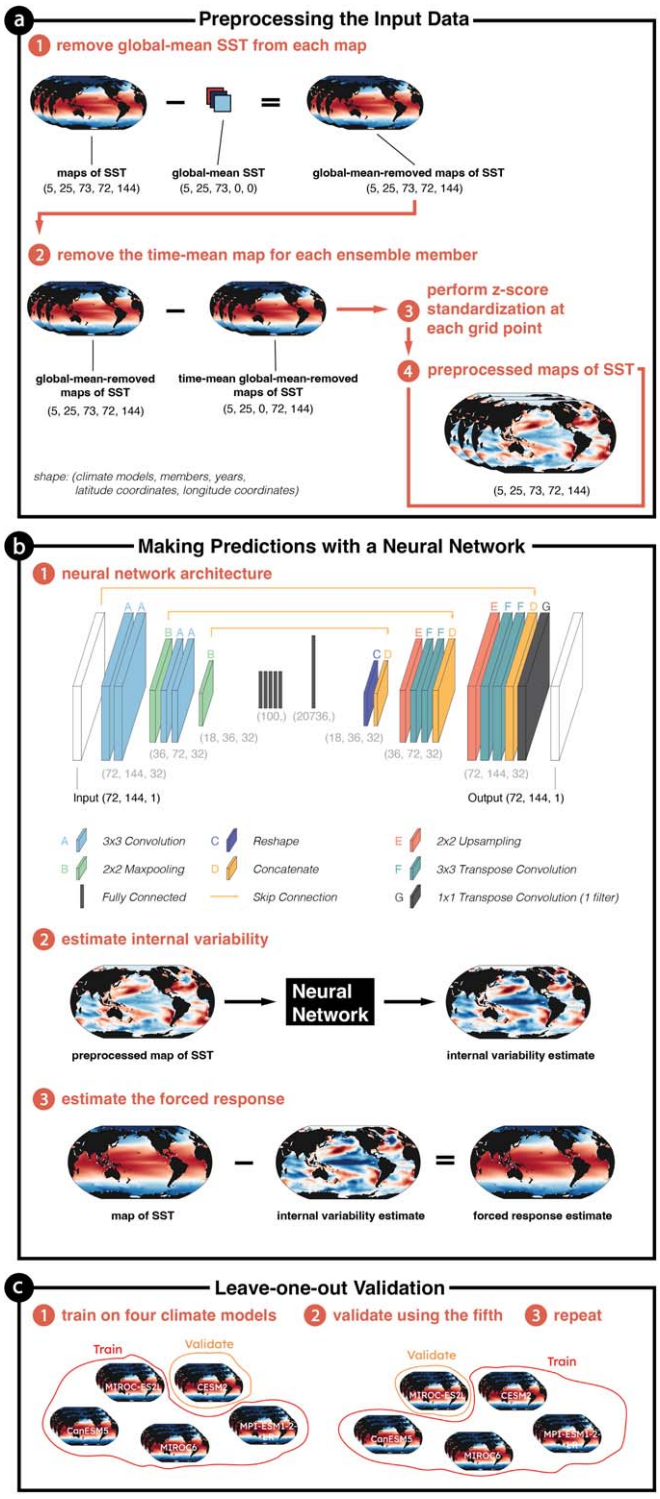


Figure 1. Visualization of the training and validation process, including (a) data preprocessing, (b) the neural network architecture and task, (c) leave-one-out validation. More information on the preprocessing procedure and the neural network architecture can be found in sections S2 and S3.

This methodology achieves three key goals for attribution of the 2023 climate. First, our method makes estimates based on the patterns within single maps of annual-mean SST, such that attribution is not dependent on knowledge of the future climate. Second, this approach does not prescribe that the forced response must evolve smoothly and can thus capture abrupt forced change. Lastly, we will demonstrate that this approach learns complex, nonlinear patterns of both the forced response and internal variability, which allows for nuanced attribution that relies on more than just the mean forced response simulated by climate models.

We evaluate the performance of our neural network using a leave-one-out approach. Neural networks are trained on five different train/validation splits, where four climate models are used for training and one is

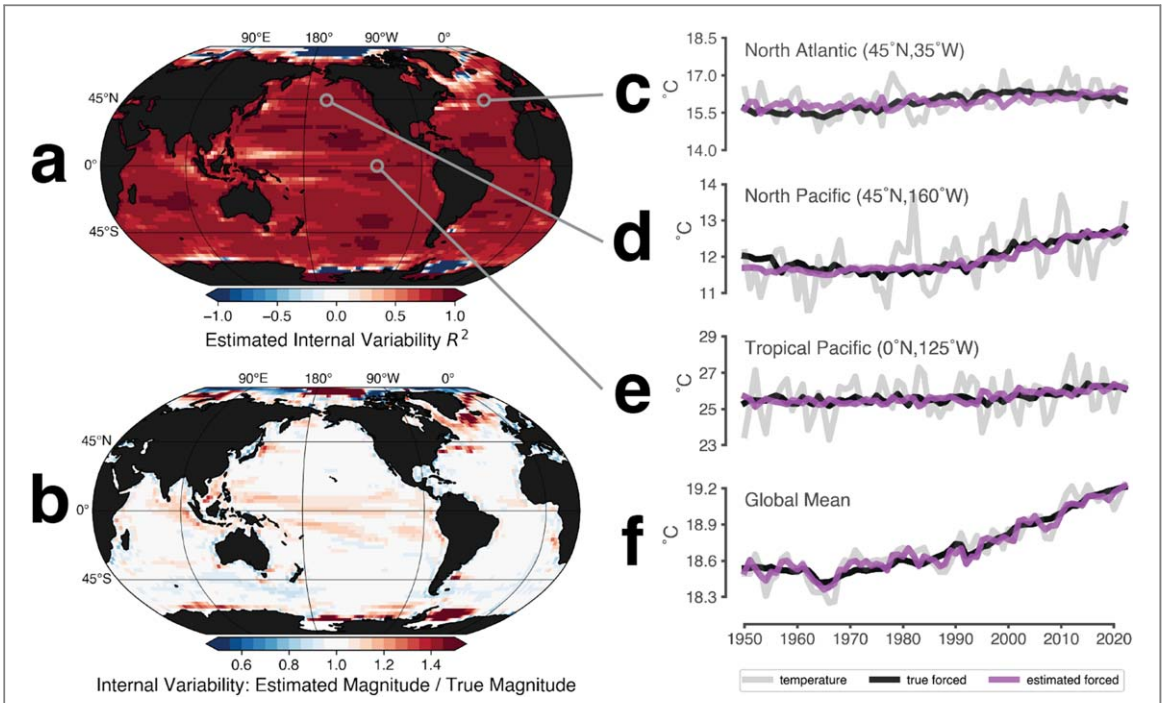


Figure 2. Performance of the neural network for separating the forced and internal components of global SST variability in climate model large ensembles. (a) R^2 score of the estimated internal variability relative to the true internal variability averaged over the climate models in the leave-one-out test sets. (b) The estimated magnitude of internal variability divided by the true magnitude of internal variability within the test sets. Ones indicate regions where the magnitude of the estimated internal variability is equal to the true magnitude of internal variability. Blues indicate regions where the magnitude of internal variability is underestimated. Reds indicate regions where the magnitude of internal variability is overestimated. Example estimates of the forced response in CESM2 for the (c) North Atlantic (45°N, 35 °W), (d) North Pacific (45°N, 160°W), (e) Tropical Pacific (0°N, 125°W), and (f) global mean.

withheld for validation (figure 1(c)). This allows us to assess our methodology on simulations of annual-mean SST that fall outside of the training data before applying this methodology to truly out-of-sample observations. We train 10 neural networks that differ only in the random seed used to initialize their weights for five train/validation splits, such that 50 neural networks are trained in total. We use the area-weighted R^2 score (coefficient of determination) to assess the skill of our internal variability estimates for simulations in the validation set, where an R^2 score of one indicates our estimates have perfect skill and an R^2 score of zero or less indicates our estimates have no skill. In addition, we compare the magnitudes of true internal variability with our estimates by calculating the standard deviation at each gridpoint for both fields.

We make our final estimate of the observed internal variability by training neural networks on data from all five climate models then performing inference on observed SST. Our observational SST product is the National Oceanic and Atmospheric Administration (NOAA) Extended Reconstructed SST version 5 (ERSSTv5) [2]. This data is regridded to a 2.5° × 2.5°, resolution to match the data in our training set. We make estimates for the years 1940–2023, a period in which the forced response signal becomes discernible in observations [42]. We make 10 different estimates of the internal component of observed SST using neural networks with different initializations. Our final estimate of the internal variability component is the mean of these 10 estimates, which we find is more skillful on out-of-sample data than the estimate of any single neural network (figure S1).

3. Results

Using neural networks trained with leave-one-out cross-validation, we find that our approach provides skillful estimates of internal variability for out-of-sample climate model data. Across most of the globe, the R^2 score is greater than 0.7, indicating that our neural network skillfully estimates internal variability (figure 2(a)). In these regions of skill, the true and estimated magnitudes of internal variability are comparable, differing by less than 20%. Exceptions in the skill of our estimates of internal variability can be found in parts of the Arctic, North Atlantic, and the Southern Ocean, regions where SST variability is dominated by sea ice extent, and in eddy-rich regions such as the Gulf Stream and Kuroshio extension.

We compare the neural network estimate of the forced response to the true forced response when CESM2 is withheld from the training set in figures 2(c)–(f). In three regions where SST is dominated by large-scale, multi-annual climate modes–the tropical Pacific, the North Pacific, and the North Atlantic–our neural networks have

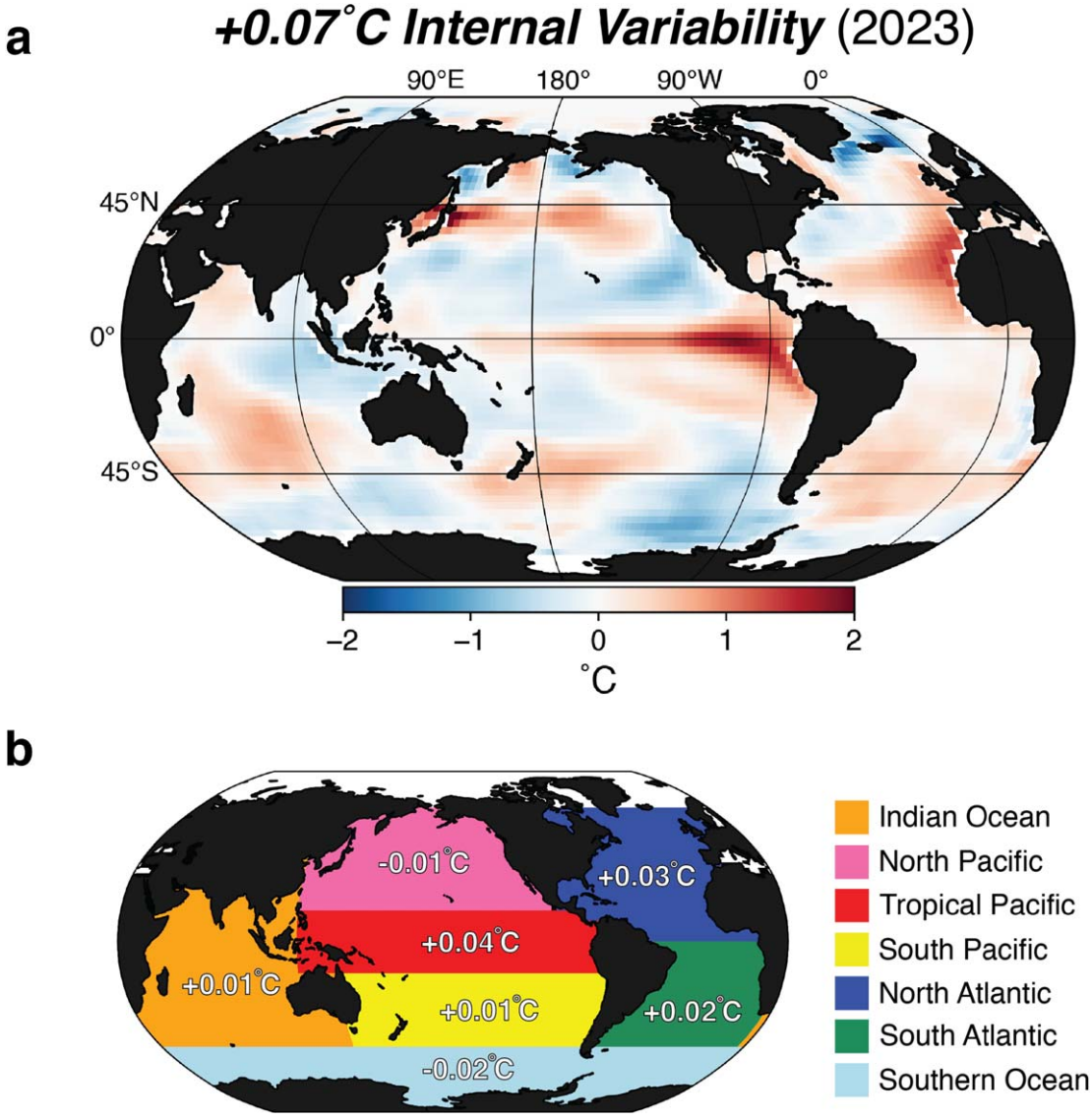


Figure 3. (a) Internal variability estimate for observed SST in 2023 attributes 0.07 °C of the global-mean SST to internal variability. (b) Regional contributions to the global-annual-mean SST anomaly due to internal variability, rounded to the nearest 0.01 °C. Due to rounding, the sum of the regional contributions appears to be +0.08 °C, despite a global-mean anomaly of +0.07 °C.

skillfully separated the internal and forced components of variability. In addition, the global mean forced response estimate closely tracks the true forced response, including the period from 1992 to 1995 when simulations of global mean SST are cool in response to the 1991 eruption of Mount Pinatubo [43]. Additional examples when other climate models were used for validation can be found in figure S2.

Using neural networks trained on all five climate model large ensembles, we now estimate how much internal variability contributed to the record-high SST observed in 2023 (figure 3). Our internal variability estimate reveals an anomalously warm tropical Pacific, contributing +0.04 °C to global-annual-mean SST. The North and South Pacific basins are both characterized by cold SST at the eastern boundaries with warm anomalies to the west, accounting for −0.01 °C and +0.01 °C of the global-annual-mean SST, respectively. Warm anomalies are widespread in the Atlantic Ocean. The largest warm anomalies are in the northern subtropics, where the North Atlantic contributes +0.03 °C, while the South Atlantic contributes +0.02 °C. The Indian Ocean presents contrasting warm anomalies in the southwest and cold anomalies in the northeast, resulting in a basin-wide contribution of +0.01 °C. The Southern Ocean accounts for −0.02 °C of the global-annual-mean SST. In total, we estimate that internal variability accounted for a +0.07 °C anomaly in global-annual-mean SST in 2023.

Estimates of the forced component reveal persistent forced warming of global-annual-mean SST over the past twenty years (figure 3(a)). The forced component in 2023 is estimated to be 0.10 °C warmer than the forced component in 2022, and 2022 is estimated to be 0.05 °C warmer than 2021, while the estimated forced response

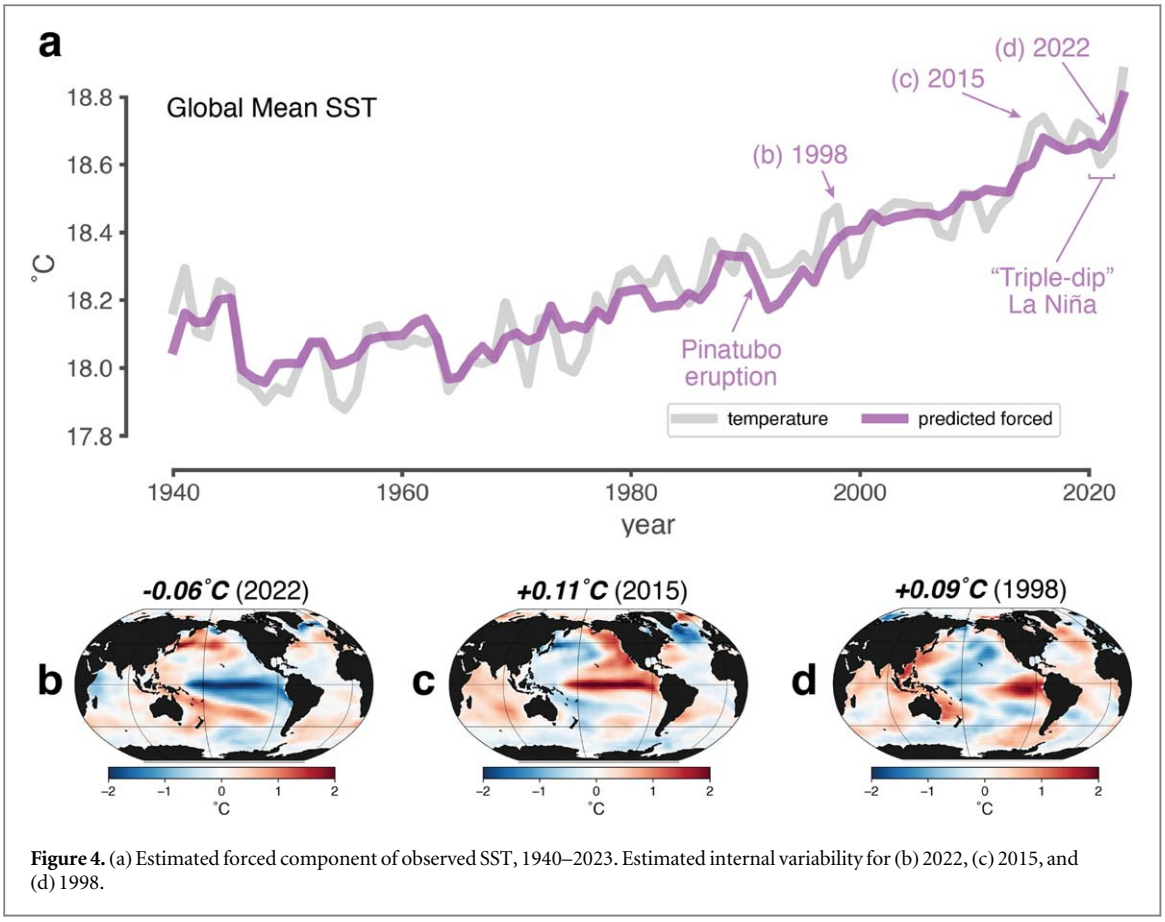


Figure 4. (a) Estimated forced component of observed SST, 1940–2023. Estimated internal variability for (b) 2022, (c) 2015, and (d) 1998.

changed by less than 0.03 °C each year from 2016–2021 (figure S3). From 2000–2023, the forced climate is estimated to have warmed by 0.4 °C, a rate of approximately 0.2 °C per decade.

4. Discussion

Our estimates of internal variability capture multi-annual, large-scale modes of variability like El Niño Southern Oscillation [ENSO; 44, 45] in the tropical Pacific, Pacific Decadal Variability [PDV; 46, 47] in the North Pacific, and Atlantic Multidecadal Variability [AMV; 48, 49] in the North Atlantic (figures 2(c)–(e)). However, these modes of internal variability may also possess characteristics that resemble the forced response [e.g., the pattern effect in the tropical Pacific and aerosol forcings in the North Atlantic; 50, 51]. Notably, our method captures the forced cooling response of SST to the 1991 eruption of Mount Pinatubo, which resembled patterns of internal variability [52]. Estimating the forced and internal components for the entire ERSSTv5 record (1940–2023) reveals forced cooling in global-annual-mean SST following the Mount Pinatubo eruption (figure 4(a)). The identification of large-scale modes of climate variability as internal variability, and the cool SST following the eruption of Mount Pinatubo as the forced response, demonstrates that neural networks are well-suited for differentiating between forced and internal patterns of change.

This study estimates that internal variability accounted for +0.07 °C of the record-high, global-mean SST in 2023. Internal variability of this magnitude does not fully explain that global-mean SST was 0.14 °C warmer than the previous record set in 2016. Performing attribution on recent years' maps of global-annual-mean SST reveals a more complete story (figure 4(a)). In the last few years, internal variability has reduced global-annual-mean SST. Internal variability in 2020, 2021, and 2022 contributed to global-mean SST anomalies of +0.03 °C, −0.05 °C, and −0.06 °C, respectively (figure S4; figure S5). These anomalies can be attributed to the cold phase of ENSO, which is a strong driver for unforced variability in global-annual-mean SST [53]. From 2020 to 2022, the climate experienced a 'triple dip' in cold-phase ENSO conditions [figure S5; 54], driving global-mean SST down and veiling the forced warming of global-mean SST (figures 4(a), (b)). Thus, the combination of continued global warming (figure 4(a); 55) and the shift to warm-phase ENSO conditions (figure 3; 56) caused the sudden increase in global-mean SST in 2023.

In addition to the warm-phase ENSO in the tropical Pacific, the South Pacific, the Atlantic Ocean, and the Indian Ocean also added to the warm anomaly in 2023 (figure 3(b)). South Pacific SST patterns resemble the

negative phase of the Interdecadal Pacific Oscillation [IPO; 57, 58] while Indian Ocean SST patterns indicate a positive-phase Indian Ocean Dipole [IOD; 59]. The Atlantic Ocean was characterized by wide-spread warmth, in part due to the warm-phase of AMV [60]. Thus, the internally-driven warm anomalies in 2023 global-mean SST were not only a product of a warm tropical Pacific, a common signature of a warm-phase ENSO event, but also concurrent warm SST anomalies due to internal variability across the oceans worldwide. Since SST anomalies in the tropical Pacific can influence SST in other basins (e.g., the North Atlantic, Central Atlantic, and Indian oceans) through atmospheric teleconnections, the widespread warmth may have been related to ENSO as well [61–63].

Such a magnitude of global-mean internal variability is not an outlier in our observational record. Our estimate that internal variability contributed +0.07 °C to 2023 global-mean SST can be compared with the conditions in 2015 and 1998 (figures 4(c)–(d)). In 2015, internal variability accounted for a +0.11 °C anomaly in global-annual-mean SST, while in 1998 internal variability accounted for +0.09 °C. Like 2023, both 2015 and 1998 featured strong warm-phase ENSO events [64], though they came in different forms 2023 featured a spring transition from a cold-phase to a warm-phase ENSO event, while 1998 experienced a transition from the warm-phase to the cold-phase. ENSO was in its warm phase across all 12 months of 2015.

Timeseries of each ocean basin's contribution to global internal variability (figure S6) reveals that the warmth in 1998 was dominated by warm contributions from the Indian Ocean (+0.06 °C), North Atlantic (+0.03 °C), and tropical Pacific (+0.03 °C), which were partially offset by a cold North Pacific (−0.02 °C). 2015 experienced an extremely warm tropical Pacific (+0.10 °C) and Indian Ocean (+0.06 °C), though global-mean SST was dampened by cold anomalies that contributed −0.02 °C to global-mean internal variability in each of the North Atlantic, South Pacific, and Southern Ocean. Climate indices associated with these anomalies can be found in figure S7. Whereas global-mean internal variability featured contributions in excess of +0.06 °C in one or more of its ocean basins in both 1998 and 2015, 2023 was unique in that the contributions from individual ocean basins were relatively modest (figure S6). Instead, the +0.07 °C global-mean internal variability was achieved through a 'perfect storm' of above-normal temperatures across most of the global ocean.

Our results suggest that abrupt forced change may have played a role in the record-breaking 2023 SST. The estimated 0.1 °C change in the global-annual-mean forced component from 2022 to 2023 is the largest such change in our observational record (figure S3), which may indicate that global warming has accelerated in recent years. Accelerated warming has been suggested by several other studies as the climate community seeks the drivers behind the recent record global-mean temperatures. One possible driver is the decrease in ship tracks due to the new shipping regulations mandated in 2020. Sulfur and aerosol emissions from ships brighten low marine clouds resulting in radiative cooling, and thus a reduction of these ship tracks could drive accelerated warming [6, 7]. Other unexpected changes to our climate system, such as reduction in low-cloud cover due to an emerging cloud feedback [8], could also be responsible for accelerated warming rates. While this method is limited in its ability to identify the exact physical driver for this abrupt change in the SST forced response, this result stands as an important line of evidence for the presence of accelerated change. This result should be used alongside other attribution methods to build confidence and consensus in estimates of global internal variability and the forced response [31].

It is necessary to mention that while 2022–2023 is the largest change in forced global-annual-mean SST in our observational record, there are three periods (1987–1988, 1996–1997, 2015–2016) that experienced comparable 0.07 °C–0.09 °C shifts. The forced response estimate changes very little from 2016 to 2021 before rapidly warming from 2021 to 2023. It is possible that the forced response of SST was concealed from 2016 to 2021, but ocean temperatures continued to rise at depth [e.g., 55]. This speaks to how difficult attribution is, and emphasizes that our results suggest a rapid change in the detectability of the forced response in maps of annual-mean maps of SST, but not necessarily an abrupt change in the forced response itself.

A novel result of this work is that the neural network can separate internal variability and the forced response when provided only a snapshot of annual-mean SST. Unlike traditional methods for detecting the forced response, such as estimating the forced response as a linear or fourth-order polynomial fit to the data, our approach does not have access to time-varying information or recent trends in the climate. This is a much harder task, and yet, this approach yields comparable skill to a fourth-order polynomial (figure S8). One benefit of this sample-by-sample method is that it can detect abrupt year-to-year changes in the forced climate since it does not make any assumptions about how the climate evolves over time. This method is also robust against the availability of new observations. Where re-fitting a fourth-order polynomial to observations following the release of 2024 data would cause the fourth-order polynomial, and thus the forced response estimate, to change, our trained neural networks provide consistent estimates.

While this method provides skillful attribution for annual-mean SST, there are several considerations for future study that may further improve its performance. Our method makes estimates of the internal variability component from annual-mean maps of SST. While using seasonal maps of SST provided similar results (figure S9), including additional fields [such as ocean heat content, precipitation or soil moisture, e.g., 26, 55]

may contain unique patterns of internal variability and forced change. However, providing the neural networks with more information is not inherently beneficial, as it may lead to overfitting, and fields like precipitation and soil moisture have poorer representation in climate models and may not generalize to observations. As with any data-driven method trained on climate model simulations, this method requires that climate model representations of annual-mean SST are consistent with observations. However, we have designed our neural network to learn patterns of both the forced response and internal variability (section S1), and thus these results are more nuanced than projecting the simulated patterns of the forced response onto observations.

5. Conclusions

Global-annual-mean SST in 2023 was the warmest on record. This unprecedented extreme has prompted questions of how much of this warming can be attributed to an abrupt change in the forced response or internal variability [e.g., 3]. Our deep-learning approach finds that internal variability contributed +0.07 °C to global-annual-mean SST in 2023, due to anomalously warm conditions across much of the global ocean, including the Pacific, Atlantic, and Indian Ocean basins. These warm anomalies, in addition to the persistent forced warming of SST, are responsible for the record high observed in 2023. However, our method also detects that the forced response of SST increased sharply between 2021 and 2023, which may indicate accelerated warming due to external forcings in recent years. 2023 global-annual-mean SST reached a level unseen in recent history, and attribution of such extremes can provide insight into the behavior of our changing climate. Given the impacts associated with abrupt forced change on human and natural systems, future climate should be closely monitored.

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Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: https://doi.org/10.5281/zenodo.15277784.

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